

Institute of Actuaries of India

Subject ST5 – Finance and Investment A

September 2016 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**i)**

a) In the classical tax system of corporation tax, company profits are taxed twice: once in the hands of the company and once in the hands of the investor.

In other words, the company pays tax on its profits, dividends are then paid out of post-tax profits and the investor is taxed on those dividends.

The investor may be subject to income tax on distributions and capital gains tax arising from increases in the share price.

[2]

b) In the imputation tax system the company has to deduct some of the tax payable by investors on distributions and pay it directly to the government. This amount can then be set off against the total corporation tax bill of the company.

That is, the tax deducted by the company is “imputed” to the shareholder who may be able to reclaim it if they are not liable to tax. If the rate at which they are liable to tax is greater than the rate imputed they may have to pay some more tax on their dividend.

[2]

ii) The process of selling and repurchasing assets to crystallize a gain in order to take advantage of existence of an annual tax-free allowance is referred as bed and breakfasting.

If the gain from selling and repurchasing is less than the annual tax-free allowance then no tax liability will be incurred. If subsequent capital gains are taxed on the basis of the repurchase price, rather than the original purchase price, the eventual tax liability will be reduced. However, any tax saving should be compared against the extra transaction costs involved. Also, tax authorities may insist on a minimum period between selling and repurchasing for the crystallization of the gain to be effective. In this case the investor will be taking a risk that the asset increases in price between the time of sale and the time of repurchase.

[3]**[7 Marks]****Solution 2:**

i) The principal aims of regulations are:

- to correct market inefficiencies and to promote efficient and orderly markets
- to protect consumers of financial products
- to maintain confidence in the financial system
- to help reduce financial crime

[2]

ii) The main forms of regulatory regimes include:

- Prescriptive regimes: Regulation can be prescriptive, with detailed rules setting out what may or may not be done.
- Freedom with publicity: Alternatively, it can involve freedom of action but with rules on publicity so that third parties are fully informed about the providers of financial services.
- Outcome-based regimes: Allow freedom of action, but prescribe the outcomes that will be tolerated and focuses on achieving suitable outcomes

[2]

iii) The main types of regulatory regimes for financial services firms include:

- Unregulated markets: Where no financial services specific regulations apply, market participants are instead subject to the normal laws of the land.
- Voluntary codes of conduct: Drawn up by the financial services industry itself.
- Self-regulation: Organised and operated by the participants in a particular market without government intervention.
- Statutory regulation: Herein a government body sets out the rules and policies.

[2]

[6 Marks]

Solution 3:

i) If C is the annual coupon and 100 be the nominal value, then the equation of value for the bond would be:

$$C \exp(-0.08 \times 1) + C \exp(-0.07 \times 2) + (100 + C) \times \exp(-0.06 \times 3) = 100$$

$$\text{Or, } C \{ \exp(-0.08 \times 1) + \exp(-0.07 \times 2) + \exp(-0.06 \times 3) \} = 100 - 100 \times \exp(-0.06 \times 3)$$

$$\text{Or, } 2.63 C = 16.47$$

$$\text{Or, } C = 6.27$$

Therefore, the coupon rate = 6.27 per 100 rupees = 6.27%

The par yield is 6.27%.

[2]

ii) A price of 92 implies futures rate of 8% *pa* convertible quarterly.

That is,

$$e^{\delta} = \left(1 + \frac{0.08}{4} \right)^4, \text{ where } \delta \text{ is the future rate}$$

$$\text{Or, } \delta = 7.9\%$$

Therefore, the 7.9% is the future rate.

Now, Forward rate = Futures rate – Convexity adjustment

$$\text{Convexity adjustment} = \frac{1}{2} \times \sigma^2 \times t_1 t_2$$

Here, $\sigma = 0.012$, $t_1 = 7$ and $t_2 = 7.25$

Therefore, convexity adjustment = 0.37%

Hence, forward rate = 7.9% - 0.37% = 7.53%.

[3]

iii) Interest rate forward and futures prices are negatively correlated with interest rates.

When the price of the underlying asset is negatively correlated with interest rates, a long forward contract will be more attractive than a similar long futures contract and forward prices will tend to be higher than futures prices.

This is because, with the increase in interest rate if the asset price decreases, then an investor with a long futures position makes an immediate loss because of the daily margining procedure of marking to market. As such loss tends to happen when interest rates are high; this loss will tend to be financed at higher than average interest rates.

Likewise, decrease in the asset price, which leads to an immediate gain to an investor with a long futures position, will tend to be invested at lower than average interest rates. In contrast an investor with a long forward position will not be affected in this way by interest rate movements.

Hence, forward prices will be higher than the corresponding futures prices. Consequently, forward rates must be lower than the corresponding futures rates, which is the case in above question.

[5]

iv) Delta of short call option = $-N(d_1)$, where

$$d_1 = \frac{\ln\left(\frac{S_t}{K}\right) + \left(r + \frac{\sigma^2}{2}\right) \times t}{\sigma \times \sqrt{t}}$$

$$\text{Or, } d_1 = \frac{\ln\left(\frac{99}{100}\right) + \left(0.05 + \frac{0.2^2}{2}\right) \times 0.5}{0.2 \times \sqrt{0.5}} = 0.21178$$

Or, $N(d_1) = 0.584$

That is, Delta of short call options = -0.584

To delta hedge this investment, the investor would have to buy 584 stocks ($=0.584 \times 1,000$).

The amount required for buying 584 stocks = $584 \times 99 = \text{Rs } 57,816$

Hence, the amount required to delta hedge these 1,000 options at the start of investment = Rs 57,816.

[4]

v) Replacing S_t and t with 99.39 and 0.4987 respectively, we get

$$d_1 = 0.23913$$

or, $N(d_1) = 0.594$

That is, new delta of short call options = -0.594

To delta hedge this investment, the investor would have to buy 594 stocks ($=0.594 \times 1,000$).

But the investor has already 584 stocks, so he would have to buy 10 new stocks at the current price of 99.39 per stock.

Therefore, the additional amount required to delta hedge this investment on the next day = $99.39 \times 10 = \text{Rs } 993.90$

[3]

[17 Marks]

Solution 4:

i) Value at Risk is a measure of the maximum loss that might be suffered on a portfolio or a holding in a specified timescale and with a specified probability.

For example, VaR for Market Risk would capture the change in the value of a portfolio due to movements in the market value of the assets held. [1]

ii) The key shortcomings of VaR are:

- VaR calculated assuming stock log returns are normally distributed may result in an underestimate, because short period stock returns have exhibited a fat-tailed distribution.
- Past experience used to calibrate the distribution may not hold true in future.
- VaR does not say anything about how bad losses could actually be.
- VaR ignores any problems relating to market liquidity.
- There is no single optimal choice for the time horizon and confidence level at which to calculate VaR making it difficult to collate and compare across different components and entities.

[2]

iii) Stress testing the solvency of pension funds and their ability to meet any promised payouts or returns in the event of fall in asset prices.

Stress testing the accumulated return of the fund's investments (accumulated balance in the individual funding account at the moment of retirement).

Stress testing the accumulated volatility of the chosen investment strategy.

Stress testing the volatility associated with the contributions (failure to contribute consistently throughout the life-cycle).

Stress testing the cost associated with one unit of pension, or necessary capital (annuitization or re-investment risk).

Stress testing longevity risks which affect the value of the pension received.

[4]

iv) The disadvantage of VaR is that it does not allow the user to investigate specific situations or combinations of situations that may result in a loss i.e. VaR only looks at past events or future events based on data from past events.

Stress testing allows the user to model specific scenarios - effect of market turmoil or identify weak areas in the portfolio. In this respect, a stress test can focus on the risk factors that the portfolio is most exposed to and could be tailored to reveal weaknesses in the portfolio.

The disadvantage of stress testing is that the user selects scenarios which adds an element of subjectivity. Whereas VaR models the likely scenarios based on what has actually happened in the past. It is therefore less subjective, though parameter estimates and choice of model may still be subjective.

[3]

[10 Marks]

Solution 5:

i) Credit default swap would provide a payment if a particular credit event occurs.

Thus, it will aim to reduce the credit risk associated with the corporate bond.

This will also help the insurance company in case it has reached its internal credit limit with a particular client, but wish to maintain their relationship with that client to reduce their aggregate exposure to the client.

However, credit default swap has its own counterparty credit risk.

Credit default swap comes at a cost.

The package of corporate bond plus CDS is relatively illiquid than a government bond.

[4]

ii) Multifactor model attempts to model the process generating investment returns as a function of several sources of systematic risk, which may be macroeconomic variables and/or company-specific factors.

Use in active management

Multifactor models can be used to estimate the appropriate expected return on a share given a set of risk factors and its estimated factor returns (the coefficients of the risk factors used in the model).

This can be done by estimating the expected values of each of the factors in the model and substituting those estimates into the model to predict the expected return on the share.

The resulting estimate would represent the return that one ought to receive from holding the share given its exposure to the various unpredictable and therefore risky factors that are deemed to influence its investment return.

If the risk factors can be predicted with greater accuracy than the market, then outperforming shares or sectors can be identified.

When the model is used to calculate expected return, this can then be compared with the expected return based on a discounted dividend or price earnings ratio model.

If the expected return indicated by the multifactor model is lower than that indicated by the current share price the share appears cheap.

Equivalently, the required rate suggested by the multifactor model can be used to discount the predicted future cashflows yielded by the security to determine at theoretical price, which can be compared with the market price.

Use in passive management

Multifactor models can also be used to identify and control the exposure of a portfolio to the different risk factors and to change the risk profile of the portfolio to better match the exposure of the liabilities.

This is because they can be used to examine the relationships between both asset and liability values and the various economic influences and other risk factors.

By quantifying the impact of the various risk factors upon both assets and liabilities multifactor models enable the investor to select the assets so that their exposure to each risk factor is similar to that of the liabilities.

Consequently, the asset and liabilities should be broadly matched, thereby reducing the investor's exposure to mismatching risk.

The same approach can also be used to determine a set of assets that replicate the expected performance of an index, if it is desired to track the performance of an index via sampling.

However, it is difficult to correctly identify the factors that affect the expected return on any particular security.

It is also problematic to estimate the relationships between those factors and the expected returns.

Further, the relationships may vary over time and consequently estimates based upon past data may not be applicable in the future.

[11]

iii) Policy loan under participating savings products would provide an alternative investment.

As the loan is secured against the policy so there is minimal credit risk.

If the policyholder doesn't repay the loan, it can be financed from the value of the policy.

However, the total loan amount should be capped so that it doesn't exceed the value of the policy.

Otherwise, policyholder would take the policy loan and wouldn't repay it.

The policy loan also helps in reducing the policy surrender rate.

The policyholders who are in need of money might prefer to take policy loan rather than to forego the whole benefits by surrendering the policy.

As this would be a new type of investment for the Company, administrative systems would require to be built.

The Company would have to decide on the interest rate it is going to charge on this type of loan.

[6]

[21 Marks]

Solution 6:

i) Inflation-linked bonds are bonds where the principal is indexed to inflation or deflation on a daily basis in terms of the official Daily CPI. They are thus designed to hedge the inflation risk of a bond. Another variant could be that the monthly/annual coupons are indexed to the historical average of daily/monthly or yearly CPI.

The real yield of any bond is the annualized growth rate, less the rate of inflation over the same period. This calculation is often difficult in the case of a normal nominal bond, because the future inflation is an unknown rate at the time of the calculation.

However, in the case of inflation-linked bonds, the bond yield is typically specified as a rate in excess of inflation, so the real yield can be easily calculated using a standard bond calculation formula.

[3]

ii) Some of the reasons for the overwhelming response could be:

- This was the first of its kind offering for retail investors
- Has an inherent natural hedge against future inflation, thereby resulting in immunisation of this risk, which otherwise can only be partially hedged using interest rate fixed to floating swaps
- Provides a reasonably predictable guidance on a real rate of return expected over the course of the investment, unlike any other asset class
- The nominal returns were lucrative and comparable to the returns investors would have otherwise made on similar fixed interest rate bonds

[3]

iii) If liabilities are linked to inflation, their present value will be sensitive to changes in inflation expectations. Hence, to hedge this sensitivity, investment should be made in assets with the same sensitivity to inflation expectations as the liabilities to reduce any mismatch in performance. Investments used to match inflation liabilities include - inflation linked bonds and inflation swaps.

The following cashflows spanning across different types of products, can be hedged effectively by investing in this inflation linked bond.

- Renewal expenses incurred towards maintaining the policy are sensitive to changes in inflation over the policy term.
- Similarly, termination expenses incurred while settling policyholder claims are sensitive to changes in inflation over the policy term.
- Since it is difficult to predict inflation with certainty and in absence of a matching investment, an additional appropriate margin for adverse deviation should be kept to avoid insufficiency of reserves in future.
- Hence, if a matching asset were available, the extent of additional prudence in reserves could be reduced, resulting in a potential reduction in reserve requirement towards this aspect. This needs to be discussed appropriately with the senior management, since this is not reduction in reserves but reduction in additional prudence which can now be hedged better.

In addition to appropriate hedging, new product categories can be explored with the availability of this inflation-linked asset class, for example:

- Index linked annuity – some trusts offer this option, however in absence of appropriate matching investments these are difficult to price and hence have been difficult to offer.
- Inflation linked increasing Sum Assured under protection plans– an inflation linked increasing risk cover would be attractive to customers. However, with availability of inflation linked assets, it has been difficult to price increasing Sum Assured products, which could now be explored.

[6]

[12 Marks]

Solution 7:

i) The ability of a company to generate profits is a vital factor in determining the market price of shares, so profit forecasting would often be the starting point for valuing shares.

The factors to be considered would be:

a) Management ability: The Company is a large life insurance company and is making profit for last 4 years so it seems that the company's management is highly regarded.

This will give investors more confidence in the prospects of this company and hence its share price would get good value.

b) Quality of products: The Company has a range of successful products. Further, its business is diversified and not dependent on one product.

This implies that the Company should be able to maintain and increase future profitability.

It is given that the Company has written diversified range of business starting from term insurance to endowment assurance (participating and non-participating) and annuity.

This diversification of business may enable to company to make profit even in the adverse situation. For example- if the mortality level increases then the term insurance written by company may lead to losses whereas the annuity may lead to profit.

Further, the split of single premium and regular premium needs to be checked. A company writing more regular premiums would have better prospects than a Company dependent on single premiums.

- c) Prospects for market growth: The Company is based in a developing nation. That is the economy is growing and the company has good prospects for growth.

The steps taken by government to increase the insurance penetration would also help the Company in growing its business.

It is given that the Company has recently launched couple of products on online channel. This would further lead to increase in sales and a scope of growth.

The Company is not overly dependent on a single distribution channels. That is, if one distribution channel fails to perform to expectation in one year, the other may be able to make up the volume.

- d) Competition: The level of competition present in the life insurance industry would have to be studied.

As the nation is developing and the current life insurance penetration is low, it implies that, in spite of competition, there would be good scope of growth.

However, this needs to be validated by a detailed study of the comparative figures of the other life insurance companies.

- e) Input costs: the extent to which the key input costs, e.g. salary, rent, overheads, etc. are expected to increase when compared to the growth of the company may have a bearing on the margins and hence overall profits.

- f) Retained profits: The Company has not paid any dividend in the last 4 years. This may imply that the company is retaining profits to finance future growth.

The availability of capital would help in writing more business in future and hence increase in business.

- g) History: The trend in performance will influence how the current performance is interpreted by the market.

As the Company's profit has been increasing over the last 4 years, this may be interpreted more favorably by the investors.

Apart from the above factors, the level of underwriting, reinsurance arrangements, claims payout history would also need to be studied.

Being a fundamental analyst, one may not have access to the detailed information to predict the future performance of the company. One would request the Company for the information which is not publicly available.

[13]

ii) It will be necessary to use all the available sources of information. The primary source is likely to be the company's published accounts but there are many other sources of information which include:

- the financial press and other commercial information providers
- the trade press
- public statements by the company
- the exchange where the securities are listed
- government sources of statutory information that a company has to provide, e.g. information provided to the regulator
- visits to the company
- discussions with company management
- discussions with competitors
- stockbrokers' publications
- credit ratings if they are available.

[2]

[15 Marks]

Solution 8:

i) The valuation of a life insurance business can be split into two distinct parts – existing in-force book as on valuation date and future new business - continue to write future new business, on a going concern basis.

Since it is impossible to have a level profit stream and hence a level dividend stream, separate components need to be estimated which could then be used to determine the profit and hence the dividend for each component.

The emergence of future profits, based on expected assumptions and projections, from the in-force book need to be estimated. The dividend expected to emerge would be one component of value ($D_{IF} = \text{dividend}_{IF} / \text{discount rate}$).

For future single block of new business written, emergence of future profits for that block on a standalone basis can be estimated based on assumptions, consistent to the ones used for estimating in-force book.

However, this value may need to be estimated for all future new business cohorts, projected together as one. The growth of future dividend would significantly depend on projected growth in future new business.

Say the growth in future is 'g' year on year, for the profits and hence dividends emerging from future new business. This could then be used to determine the second component of value ($D_{NB} = \text{dividend}_{NB} / (\text{discount rate} - g)$).

Depending on the stage of the life insurance company, the relevant proportions of the two components could vary, with a growing company having a high weightage towards D_{NB} and an established long standing company a higher weightage towards D_{IF} .

[7]

ii) From the working and formulae in the previous response, it is evident that the critical assumptions would be growth (g) and risk discount rate (r) given their combined gearing effect on the total market value of the company.

Further, the other critical assumptions that impact DCF workings would be persistency and expenses, given the long term nature of the life insurance business.

[2]

iii) Ratios related to expenses:

Expense ratio = expenses incurred / premiums collected

New business acquisition costs = acquisition expense / new business premium

While the former gives an indication of the total expenditure, the latter gives an indication on the money spent to get an additional 1 rupee of new business. Hence, an increasing cost of goods sold, should lead to an increase in premium, if not will adversely impact profitability.

Persistency ratio:

Given the long term nature of the life insurance, greater the duration for which the policyholder stays with the company, the high initial acquisition expenses can be spread over a longer time period, thereby resulting in a better value proposition for the policyholder and hence the company.

This may be compared to an adverse impact due to an acceleration of depreciation of a fixed asset acquired by a FMCG company.

Claim ratios:

Any adverse mortality or morbidity experience could impact the expected profitability of the insurer and hence needs to be monitored closely.

This is comparable to manufacturing scenario wherein a product is conceived assuming certain cost of goods, but when it goes into production the actual incurred costs are higher, thereby adversely impacting the profitability.

[3]

[12 Marks]
